

DAY-3

❖ **Day 3: DNS Server Configuration Theory (30 min):**How DNS resolves domain names to Ips **Practical (30 min):**Configure a DNS server and test domain resolution **Task:** Document the configuration and troubleshooting steps.

[Q-1] Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

[ANS]

➤ **Objective**

- Install a DNS server and create a DNS zone named **musicmania.in**.

➤ **Configuration Steps (BIND on Linux)**

➤ **Step 1: Install BIND**

- sudo apt update
- sudo apt install bind9 bind9utils bind9-doc

➤ **Step 2: Create a Zone Entry**

- sudo nano /etc/bind/named.conf.local

▪ **Add:**

```
zone "musicmania.in" {  
    type master;  
    file "/etc/bind/db.musicmania.in";  
};
```

➤ **Step 3: Create the Zone File**

- sudo cp /etc/bind/db.local /etc/bind/db.musicmania.in
Edit:
sudo nano /etc/bind/db.musicmania.in
Example configuration:
\$TTL 604800
@ IN SOA ns.musicmania.in. admin.musicmania.in. (
 2
 604800
 86400
 2419200
 604800)

@ IN NS ns.musicmania.in.
ns IN A 192.168.1.50
www IN A 192.168.1.50

➤ **Step 4: Restart DNS Service**

- `sudo systemctl restart bind9`

```
student@ubuntu:~$ sudo systemctl restart bind9
student@ubuntu:~$ sudo systemctl status bind9
● bind9.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 29 May 2024 10:45:12 AM IST; 3s ago
     Docs: man:named(8)
   Main PID: 1152 (named)
    Tasks: 6 (limit: 2271)
   Memory: 12.6M
      CPU: 58ms
   CGroup: /system.slice/bind9.service
           └─1152 /usr/sbin/named -f -u bind

May 29 10:45:12 ubuntu systemd[1]: Starting BIND Domain Name Server...
May 29 10:45:12 ubuntu systemd[1]: Started BIND Domain Name Server.
student@ubuntu:~$
```

[Q-2] Add an A record in your DNS server configuration to map `www.musicmania.in` to the IP address `192.168.1.50`, then use the `nslookup` or `dig` command to verify the mapping works.

[ANS]

➤ **Add an A Record and Verify**

➤ **A Record**

- `www` IN A `192.168.1.50`

➤ **Verify Using nslookup**

- `nslookup www.musicmania.in`

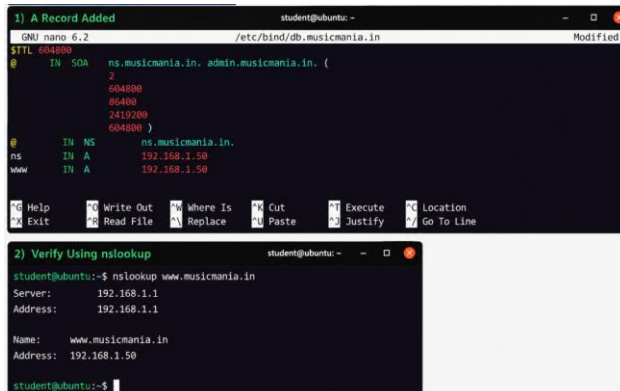
➤ **Sample Output**

- Server: 192.168.1.1
Address: 192.168.1.1

Name: `www.musicmania.in`
Address: `192.168.1.50`

➤ **Explanation**

- The A record maps the hostname www.musicmania.in to the IP address **192.168.1.50**. When users enter the domain name, the DNS server returns this IP address.



The screenshot shows two terminal windows. The top window, titled '1) A Record Added', shows the contents of the file `/etc/bind/db.musicmania.in` after editing with nano. It contains an SOA record for `ns.musicmania.in` and an A record for `www` pointing to `192.168.1.50`. The bottom window, titled '2) Verify Using nslookup', shows the output of the `nslookup www.musicmania.in` command, which returns the server address `192.168.1.1` and the target address `192.168.1.50`.

[Q-3] Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

[ANS]

➤ **Add a CNAME Record**

➤ **Configuration**

- Add the following line to the zone file:
- `play IN CNAME www.musicmania.in.`

➤ **Restart DNS**

- `sudo systemctl restart bind9`

➤ **Verify**

- `nslookup play.musicmania.in`

➤ **Sample Output**

- Name: play.musicmania.in
Canonical name: www.musicmania.in

Name: www.musicmania.in
Address: 192.168.1.50

➤ **Explanation**

- A CNAME (Canonical Name) record creates an alias. Here, **play.musicmania.in** points to www.musicmania.in, which then resolves to **192.168.1.50**.

```
1) Add CNAME Record (Edit Zone File)
student@ubuntu: ~
GNU nano 6.2 /etc/bind/db.musicmania.in Modified
$TTL 604800
@      IN      SOA     ns.musicmania.in. admin.musicmania.in. (
        2
        604800
        86400
        2419200
        604800 )
@      IN      NS      ns.musicmania.in.
ns     IN      A       192.168.1.50
www    IN      A       192.168.1.50
play   IN      CNAME   www.musicmania.in.

2) Restart DNS
student@ubuntu: ~
student@ubuntu:~$ sudo systemctl restart bind9
student@ubuntu:~$ systemctl status bind9
● bind9.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed May 29 11:05:21 IST 2024; 3s ago
     Docs: man:named(8)

3) Verify
student@ubuntu:~$ nslookup play.musicmania.in
Server:      192.168.1.1
Address:     192.168.1.1#53

Name:       play.musicmania.in
Canonical name: www.musicmania.in
Name:       www.musicmania.in
Address:    192.168.1.50
student@ubuntu:~$
```

**[Q-4] Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

Hint: Include at least three different troubleshooting commands or checks in your documentation.**

[ANS]

➤ **Problem**

The domain **musicmania.in** is not resolving correctly.

➤ **Step-by-Step Troubleshooting**

➤ **Step 1: Check DNS Configuration**

- Verify that the zone configuration is correct.
- Command:
- `sudo named-checkconf`
- Expected Result:
- No errors should be displayed.

➤ **Step 2: Verify Zone File**

- Check for syntax errors in the zone file.
- Command:
- `sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in`
- Expected Result:
- OK

➤ **Step 3: Restart the DNS Service**

- After making changes, restart BIND.
- Command:
- `sudo systemctl restart bind9`
- Check service status:
- `sudo systemctl status bind9`
- Expected Result:
- The service should show **active (running)**.

➤ **Step 4: Test Domain Resolution**

- Use **nslookup**.
- `nslookup www.musicmania.in`
- or use **dig**.
- `dig www.musicmania.in`
- Expected Result:
- `www.musicmania.in`
Address: 192.168.1.50

➤ **Step 5: Check Log Files**

- If the problem continues, inspect DNS logs.
- `sudo journalctl -u bind9`
- or
- `sudo tail -f /var/log/syslog`
- Look for errors such as:
- Zone file syntax error
- Permission denied
- Service failed to start

➤ Step 6: Verify Firewall (Optional)

- Ensure DNS traffic is allowed.
- `sudo ufw status`
- If necessary:
- `sudo ufw allow 53`

1) Check DNS Configuration <pre>student@ubuntu:~\$ sudo named-checkconf</pre> <pre>student@ubuntu:~\$</pre> Expected Result: No errors should be displayed.	2) Verify Zone File <pre>student@ubuntu:~\$ sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in</pre> <pre>zone musicmania.in/IN: loaded serial 2</pre> <pre>OK</pre> <pre>student@ubuntu:~\$</pre> Expected Result: OK
3) Restart the DNS Service & Check Status <pre>student@ubuntu:~\$ sudo systemctl restart bind9</pre> <pre>student@ubuntu:~\$ sudo systemctl status bind9</pre> <pre>● bind9.service - BIND Domain Name Server</pre> <pre>Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)</pre> <pre>Active: active (running) since Wed May 29 11:15:45 IST 2024; 2s ago</pre> <pre>Docs: man:named(8)</pre> <pre>Main PID: 1234 (named)</pre> <pre>Tasks: 6 (limit: 2277)</pre> <pre>Memory: 12.1M</pre> <pre>CGroup: /system.slice/bind9.service</pre> <pre>└─1234 /usr/sbin/named -f -u bind</pre> <pre>student@ubuntu:~\$</pre> Expected Result: The service should show active (running).	4) Test Domain Resolution (nslookup & dig) <pre>student@ubuntu:~\$ nslookup www.musicmania.in</pre> <pre>Server: 192.168.1.1</pre> <pre>Address: 192.168.1.1#53</pre> <pre>Name: www.musicmania.in</pre> <pre>Address: 192.168.1.50</pre> <pre>student@ubuntu:~\$</pre> <pre>student@ubuntu:~\$ dig www.musicmania.in</pre> <pre>; <<>> DiG 9.18.1-1ubuntu1.1-Ubuntu <<>> www.musicmania.in</pre> <pre>;; global options: +cmd</pre> <pre>;; Got answer:</pre> <pre>;;->HEADER<-- opcode: QUERY, status: NOERROR, id: 12345</pre> <pre>;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 1</pre> <pre>;; QUESTION SECTION:</pre> <pre>;www.musicmania.in. IN A</pre> <pre>;; ANSWER SECTION:</pre> <pre>www.musicmania.in. 604800 IN A 192.168.1.50</pre> <pre>;; SERVER: 192.168.1.1#53(192.168.1.1)</pre> <pre>;; WHEN: Wed May 29 11:16:30 IST 2024</pre> <pre>;; MSG SIZE rcvd: 81</pre> <pre>student@ubuntu:~\$</pre> Expected Result: www.musicmania.in Address: 192.168.1.50
5) Check Log Files <pre>student@ubuntu:~\$ sudo journalctl -u bind9 -n 10</pre> <pre>May 29 11:15:45 ubuntu named[1234]: zone musicmania.in/IN: loaded serial 2</pre> <pre>May 29 11:15:45 ubuntu named[1234]: all zones loaded</pre> <pre>May 29 11:15:45 ubuntu named[1234]: running</pre> <pre>May 29 11:15:45 ubuntu systemd[1]: Started BIND Domain Name Server.</pre> <pre>student@ubuntu:~\$</pre> <pre>student@ubuntu:~\$ sudo tail -f /var/log/syslog</pre> <pre>May 29 11:16:02 ubuntu named[1234]: client 192.168.1.100#54321: query:</pre> <pre>www.musicmania.in IN A +E (192.168.1.1)</pre> <pre>May 29 11:16:02 ubuntu named[1234]: client 192.168.1.100#54321: query:</pre> <pre>www.musicmania.in IN A +E (192.168.1.1)</pre> <pre>student@ubuntu:~\$</pre> Look for errors such as: • Zone file syntax error • Permission denied • Service failed to start	6) Verify Firewall (Optional) <pre>student@ubuntu:~\$ sudo ufw status</pre> <pre>Status: active</pre> <pre>To Action From</pre> <pre>--</pre> <pre>53/tcp ALLOW Anywhere</pre> <pre>53/udp ALLOW Anywhere</pre> <pre>22/tcp ALLOW Anywhere</pre> <pre>22/tcp (v6) ALLOW Anywhere (v6)</pre> <pre>53 (v6) ALLOW Anywhere (v6)</pre> <pre>53/udp (v6) ALLOW Anywhere (v6)</pre> <pre>22/tcp (v6) ALLOW Anywhere (v6)</pre> <pre>student@ubuntu:~\$</pre> OR <pre>student@ubuntu:~\$ sudo ufw allow 53</pre> <pre>Rules updated (v6)</pre> <pre>student@ubuntu:~\$ sudo ufw status</pre> <pre>Status: active</pre> <pre>To Action From</pre> <pre>--</pre> <pre>53/tcp ALLOW Anywhere</pre> <pre>53/udp ALLOW Anywhere</pre> <pre>22/tcp ALLOW Anywhere</pre> <pre>student@ubuntu:~\$</pre>